On Learning Underlying Forms

Adam Jardine



CUNY Colloquium
October 22, 2026

Johor Malay (Onn, 1976)

SR	**gloss**
[pəŋarah]	NOM-‘direct’
[pəmbawə]	NOM-‘carry’
[pəndaki]	NOM-‘climb’
[pəŋgali]	NOM-‘dig’

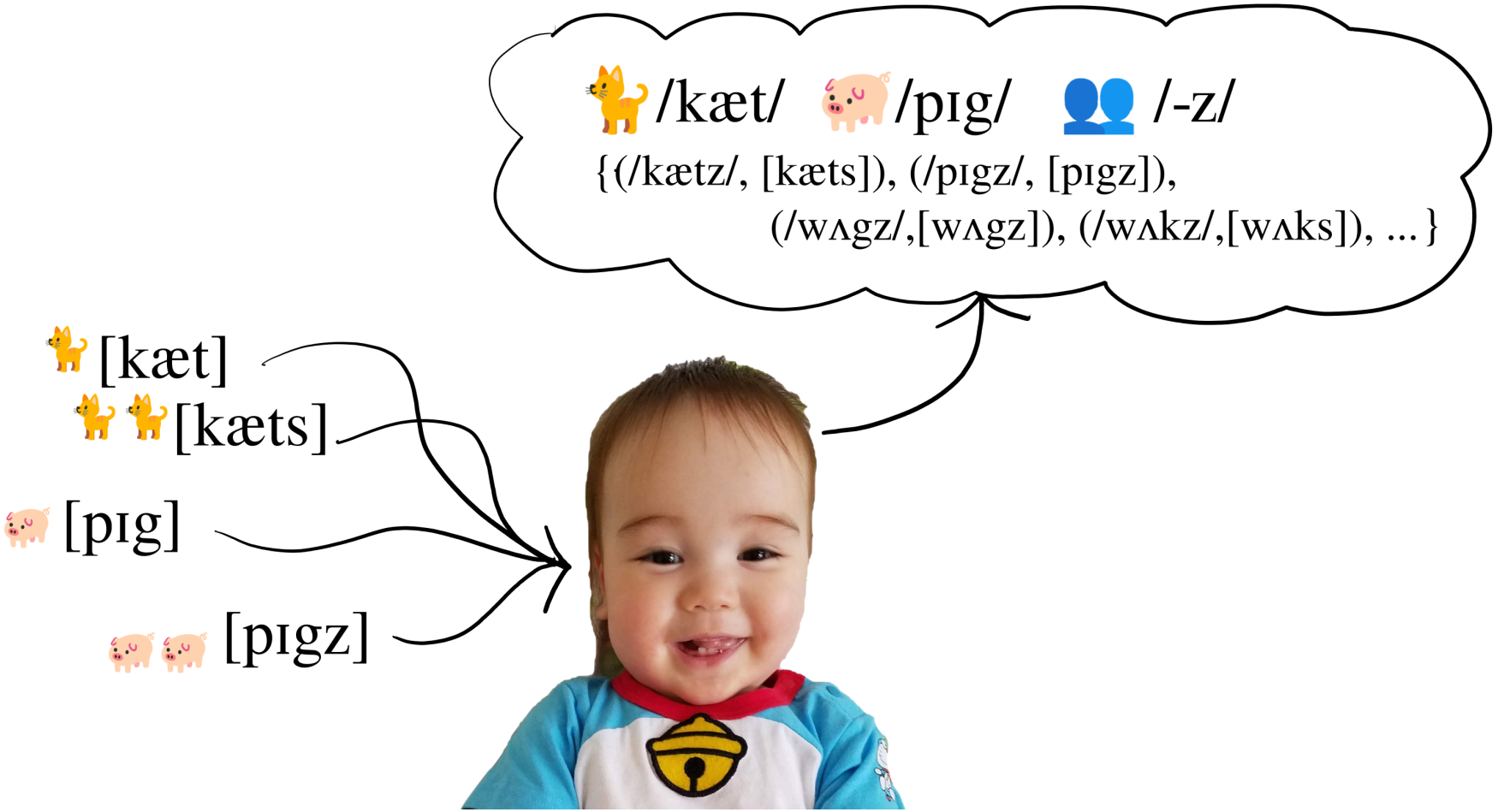
Introduction

$$\overbrace{\qquad\qquad\qquad}^{\lambda}$$

NOM	$\mapsto$	/pəŋ/
CARRY	$\mapsto$	/bawa/
CLIMB	$\mapsto$	/daki/
$\vdots$	$\vdots$	$\vdots$
NOMDIRECT	$\mapsto$	/pəŋarah/
NOMCARRY	$\mapsto$	/pəŋbawa/
NOMCLIMB	$\mapsto$	/pəŋdaki/
$\vdots$	$\vdots$	$\vdots$

Introduction

		$\overbrace{\quad\quad\quad\quad}^{\lambda}$		$\overbrace{\quad\quad\quad\quad}^{\varphi}$	
NOM	$\mapsto$	/pəŋ/	/pəŋ/	$\mapsto$	[pəŋ]
CARRY	$\mapsto$	/bawa/	/bawa/	$\mapsto$	[bawə]
CLIMB	$\mapsto$	/daki/	/daki/	$\mapsto$	[daki]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
NOMDIRECT	$\mapsto$	/pəŋarah/	/pəŋarah/	$\mapsto$	[pəŋarah]
NOMCARRY	$\mapsto$	/pəŋbawa/	/pəŋbawa/	$\mapsto$	[pəŋbawə]
NOMCLIMB	$\mapsto$	/pəŋdaki/	/pəŋdaki/	$\mapsto$	[pəŋdaki]
$\vdots$	$\vdots$	$\vdots$	/əŋbə/	$\mapsto$	[əmbə]
			$\vdots$	$\vdots$	$\vdots$


/kæt/ /pɪg/ /-z/
{(/kætz/, [kæts]), (/pɪgz/, [pɪgz]),
(/wʌgz/,[wʌgz]), (/wʌkz/,[wʌks]), ... }
[kæt]
[kæts]
[pɪg]
[pɪgz]

Target

λ			φ		
NOM	↦	/pəŋ/	/pəŋ/	↦	[pəŋ]
CARRY	↦	/bawa/	/bawa/	↦	[bawə]
CLIMB	↦	/daki/	/daki/	↦	[daki]
⋮	⋮	⋮	⋮	⋮	⋮
NOMDIRECT	↦	/pəŋarah/	/pəŋarah/	↦	[pəŋarah]
NOMCARRY	↦	/pəŋbawa/	/pəŋbawa/	↦	[pəŋbawə]
NOMCLIMB	↦	/pəŋdaki/	/pəŋdaki/	↦	[pəŋdaki]
⋮	⋮	⋮	/əŋbə/	↦	[əmbə]
			⋮	⋮	⋮

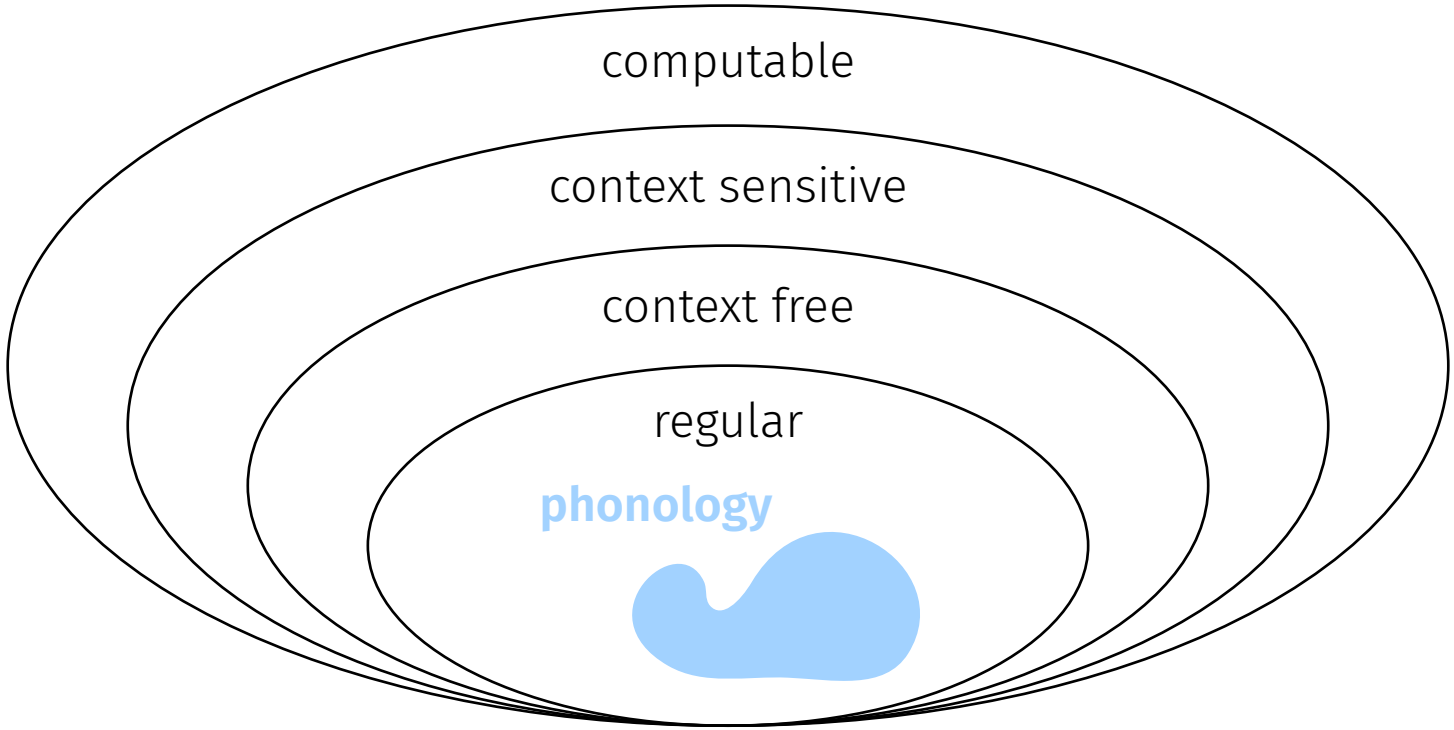
Input

Finite sample of $\varphi \circ \lambda$

(DIRECT,	[pəŋarah])
(CARRY,	[pəmbawə])
⋮	⋮
(NOMDIRECT,	[pəŋarah])
(NOMCARRY,	[pəmbawə])
(NOMCLIMB,	[pəndaki])
(NOMDIG,	[pəŋgali])

The learning problem: From a set of meanings and surface forms exhibiting morphophonological alternations, return a set of underlying representations (URs) and a phonological grammar.

The path forward: What kind of *function* is the phonology?


computable
context sensitive
context free
regular
phonology

Chandlee & Jardine (2025): Dataset from Johore Malay (Onn, 1976)

- **Input**: 14 stems × 5 affixes
- **Output**: Correct lexicon and grammar representing five processes
  - Including counter-feeding
- Generalizes theoretical results of Hua et al. (2021), Hua & Jardine (2021)
- **Inductive principles**:
  - Phonological functions are **input-local**
  - Phonological functions refer to **natural classes** (in the environment)

Collaborators and Acknowledgements

- This talk based on Chandlee & Jardine (2025)
- NSF Grant #2416184 with





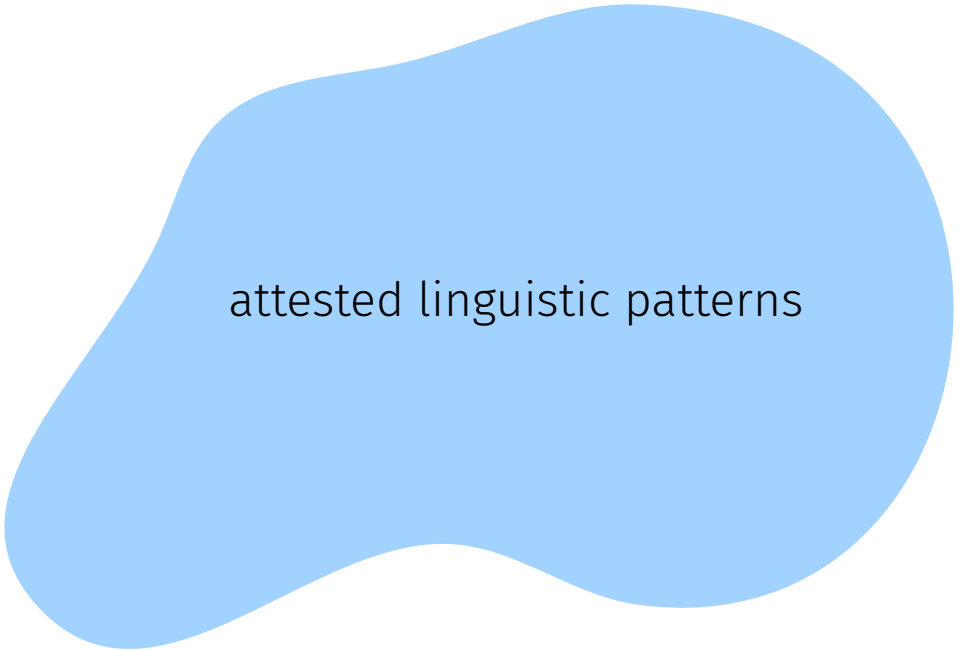


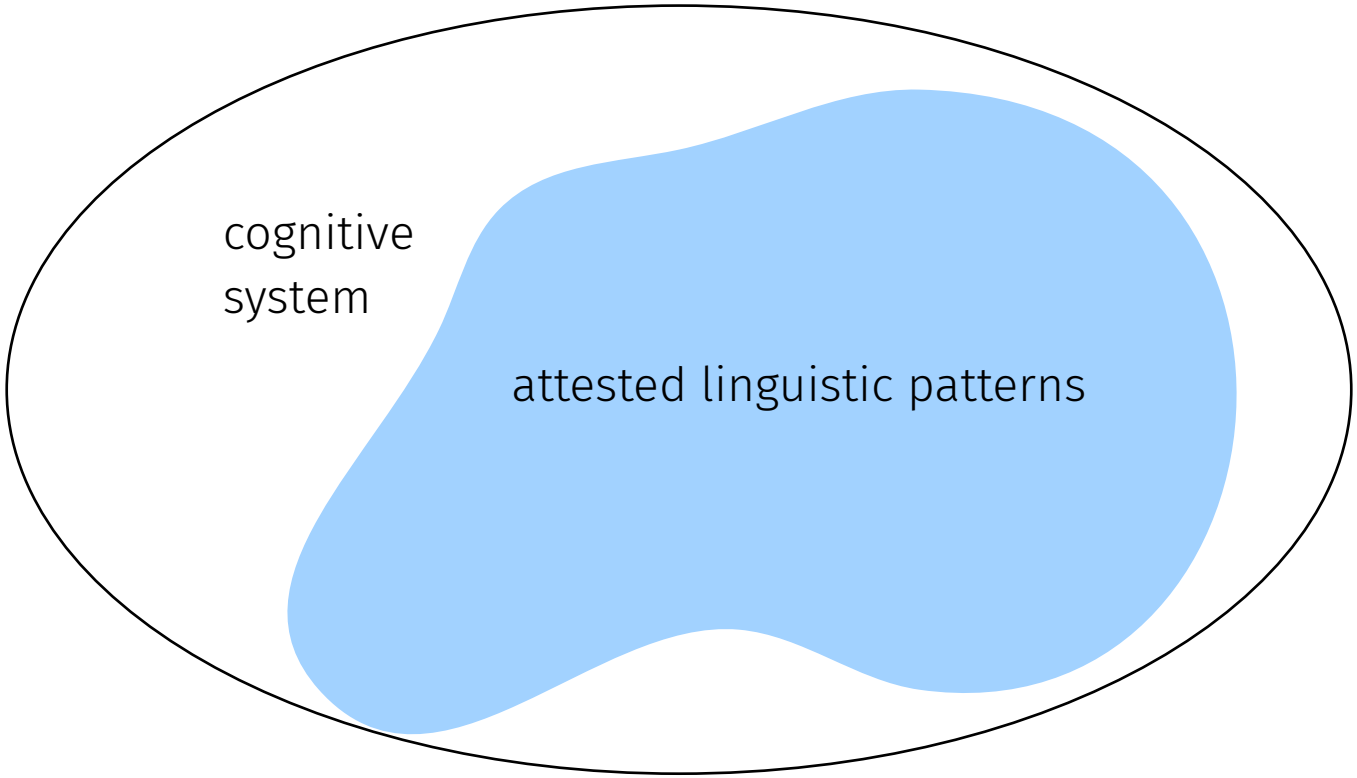
Jane Chandlee (Haverford) **Jeffrey Heinz** (Stony Brook)

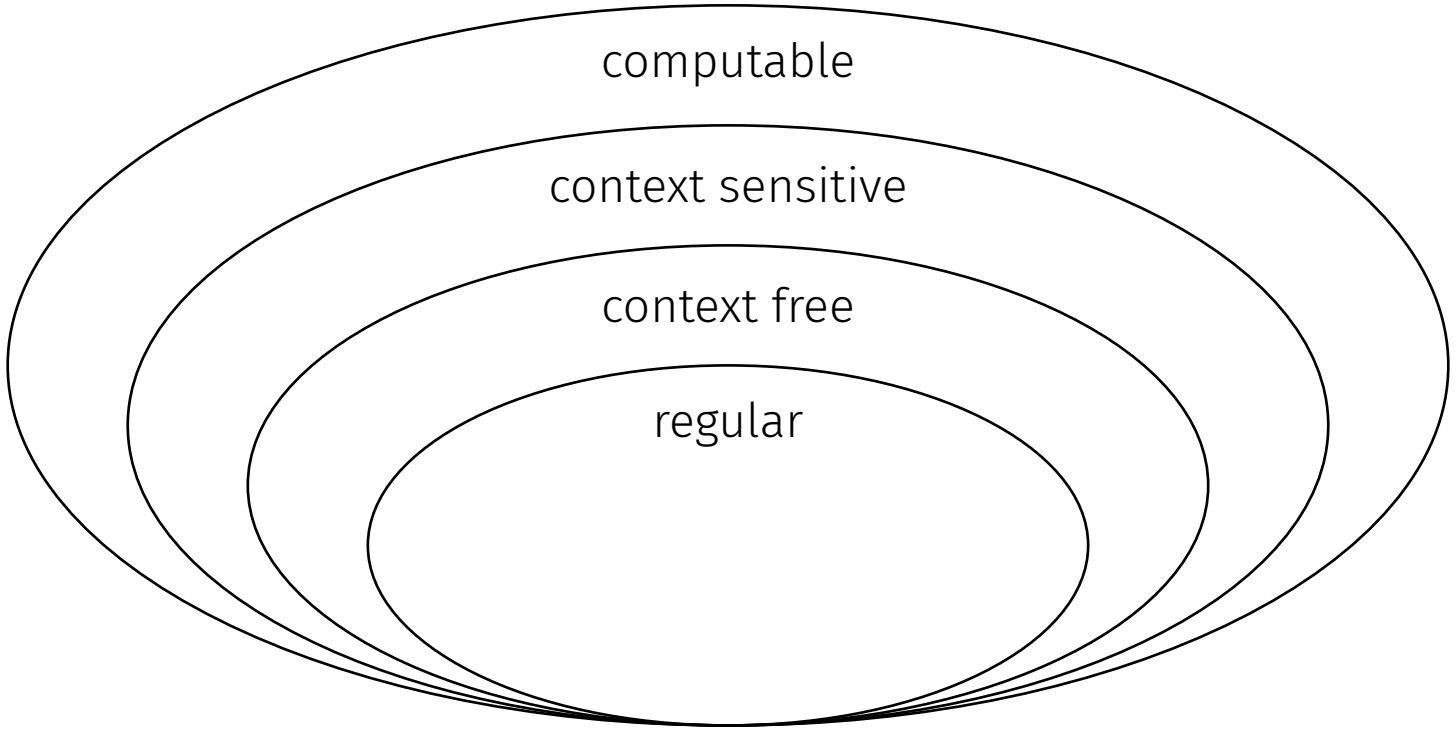
- **Grads**: Vincent Czarnecki Chenli Wang Eli Franz Zachary Helfand
  Hyunjung Joo (now at Hanyang Univ.)
  Huteng Dai (now at Univ. of Michigan)
  Wenyue Hua (Ling MA, CS PhD, now at Microsoft Research)

  UGs: Ezra Anthony Neha Jeyaram Manish Kandepalli
  Ben Evans (now at Stony Brook)

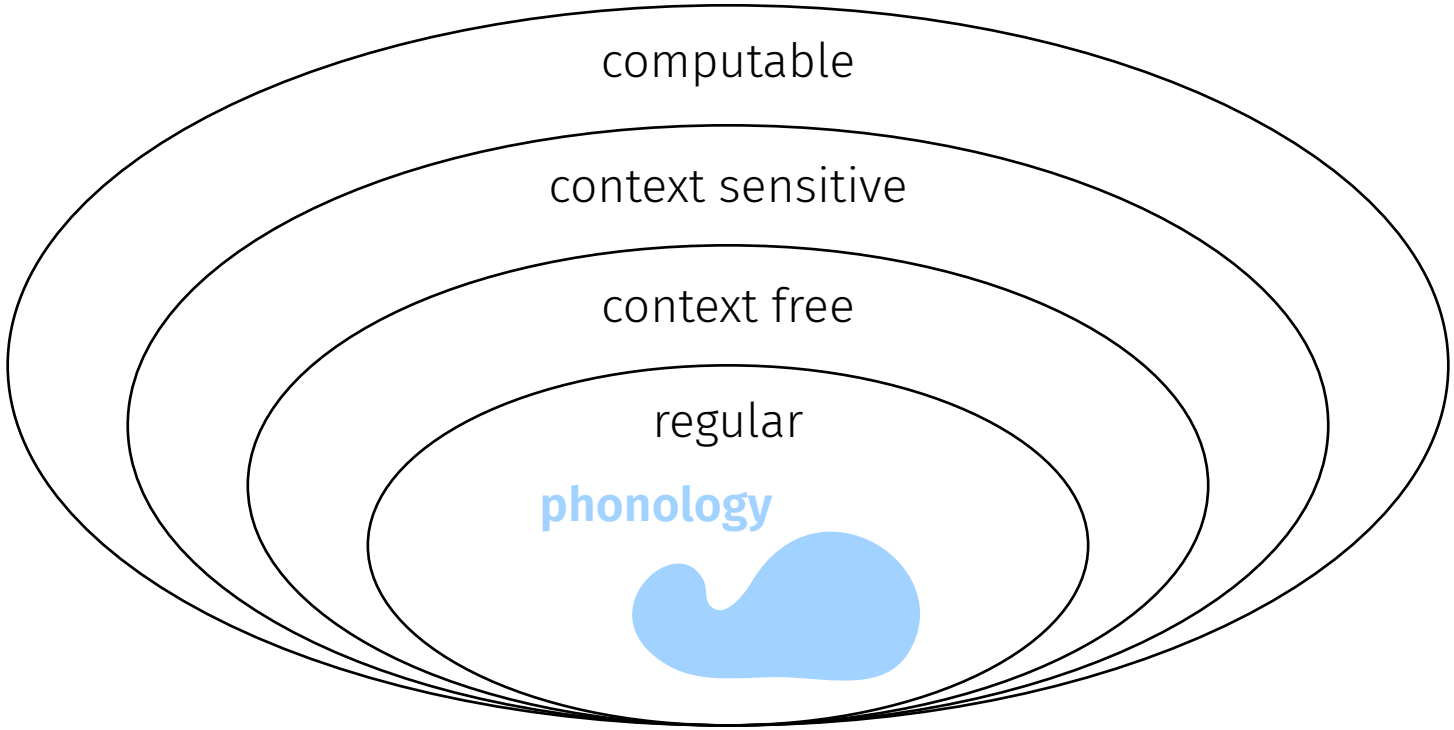
Phonology, linguistic universals, and computation


attested linguistic patterns

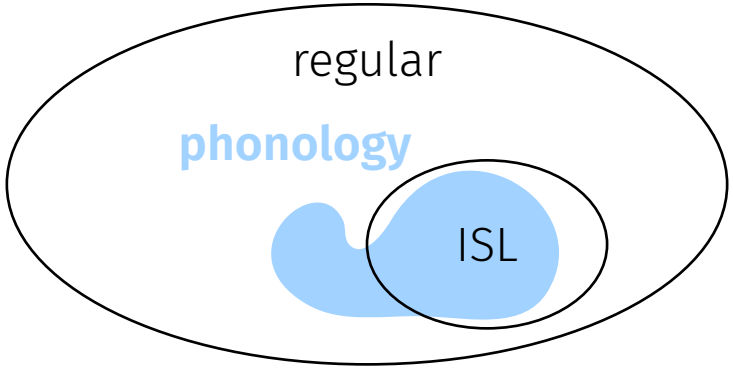

cognitive
system
attested linguistic patterns



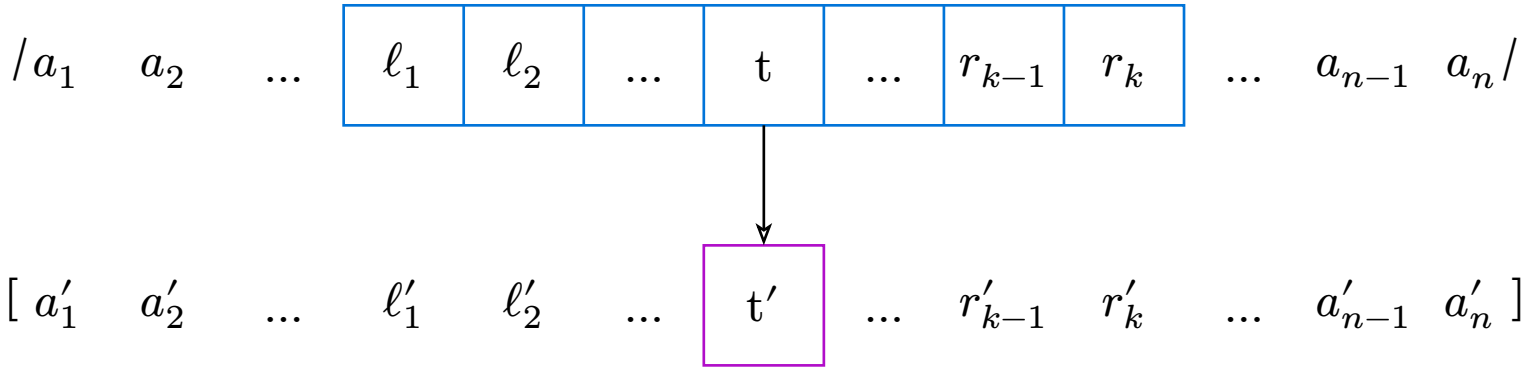
The Chomsky-Schützenberger Hierarchy (Chomsky 1957, Chomsky and Schützenberger 1963)



The Chomsky-Schützenberger Hierarchy (Chomsky 1957, Chomsky and Schützenberger 1963)
(Johnson 1972, Kaplan and Kay 1994, Heinz 2007, et seq.)



Input Strictly Local (ISL) functions
(Chandlee 2014, Chandlee and Heinz 2018)



Computation of an ISL_k function

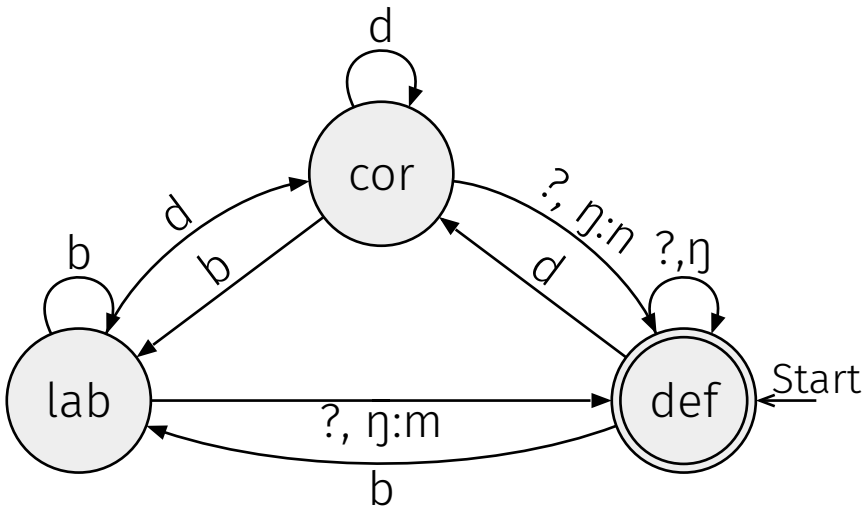
Nasal place assimilation in Johor Malay (Onn, 1976)

UR	**SR**	**gloss**
/pəŋarah/	[pəŋarah]	NOM-'direct'
/pəŋbawə/	[pəmbawə]	NOM-'carry'
/pəŋdaki/	[pəndaki]	NOM-'climb'
/pəŋgali/	[pəŋgali]	NOM-'dig'

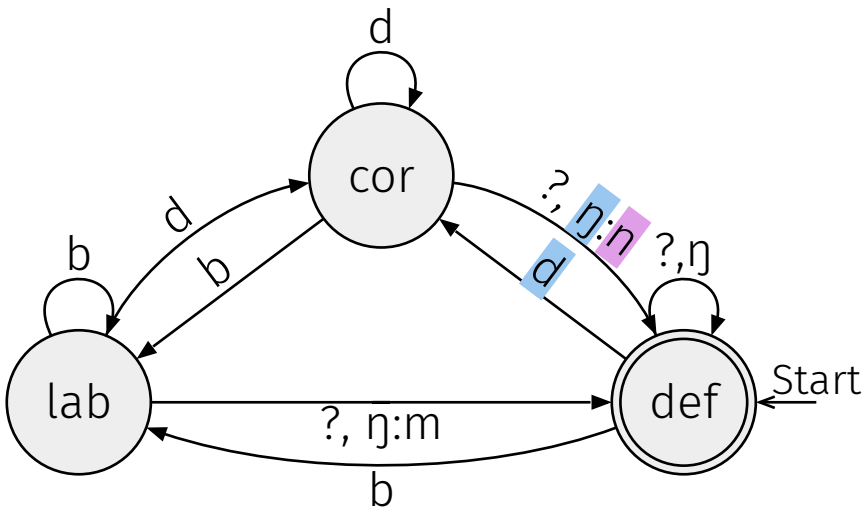
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/pəŋarah/	[pəŋarah]	NOM-‘direct’
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/pəŋgali/	[pəŋgali]	NOM-‘dig’

(partial) **finite-state transducer (FST)** for nasal place assimilation

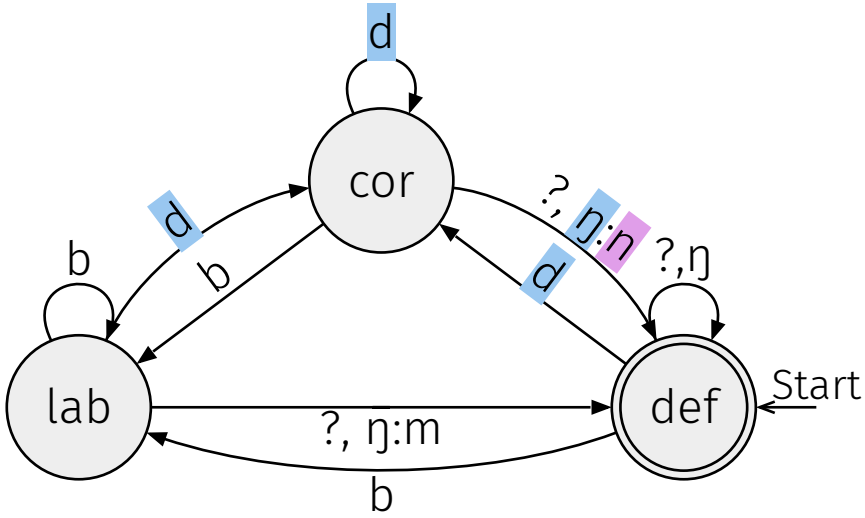


(partial) **finite-state transducer (FST)** for nasal place assimilation

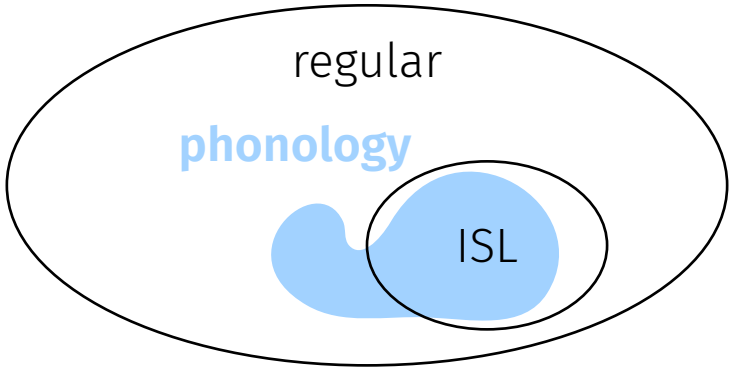


/pəŋdaki/ ↦ [pəndaki]

(partial) **finite-state transducer (FST)** for nasal place assimilation



/pəŋdaki/ ↦ [pəndaki]



95% of P-Base (Mielke 2008) is ISL
(Chandlee 2014, Chandlee and Heinz 2018)

A proposal

A proposal

Target

λ			φ		
NOM	↦	/pəŋ/	/pəŋ/	↦	[pəŋ]
CARRY	↦	/bawa/	/bawa/	↦	[bawə]
CLIMB	↦	/daki/	/daki/	↦	[daki]
⋮	⋮	⋮	⋮	⋮	⋮
NOMDIRECT	↦	/pəŋarah/	/pəŋarah/	↦	[pəŋarah]
NOMCARRY	↦	/pəŋbawa/	/pəŋbawa/	↦	[pəŋbawə]
NOMCLIMB	↦	/pəŋdaki/	/pəŋdaki/	↦	[pəŋdaki]
⋮	⋮	⋮	/əŋbə/	↦	[əmbə]
			⋮	⋮	⋮

Input

Finite sample of $\varphi \circ \lambda$

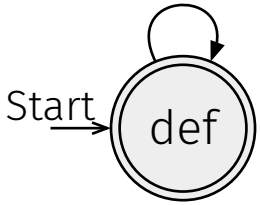
(DIRECT,	[pəŋarah])
(CARRY,	[pəmbawə])
⋮	⋮
(NOMDIRECT,	[pəŋarah])
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(NOMDIG,	[pəŋgali])

A proposal

λ

NOM	$\mapsto$	/pəŋ/
CARRY	$\mapsto$	/bawa/
CLIMB	$\mapsto$	/daki/
⋮	⋮	⋮
NOMDIRECT	$\mapsto$	/pəŋarah/
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NOMCLIMB	$\mapsto$	/pəŋdaki/
⋮	⋮	⋮

NOM:pəŋ, CARRY:bawa, CLIMB:daki, DIG:gali

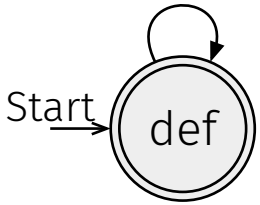


Finite-state machine for λ

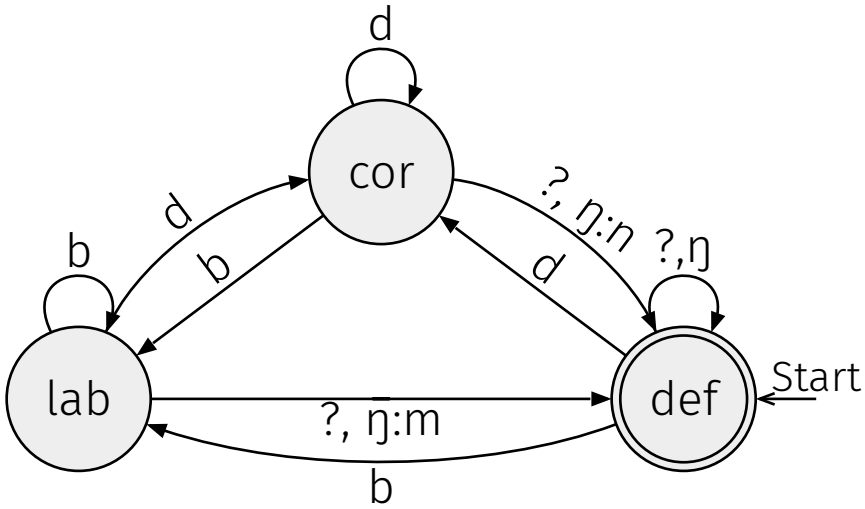
- (Hua & Jardine, 2021), (Hua et al., 2021): λ is a **homomorphism**

A proposal

NOM:pəŋ, CARRY:bawa, CLIMB:daki, DIG:gali

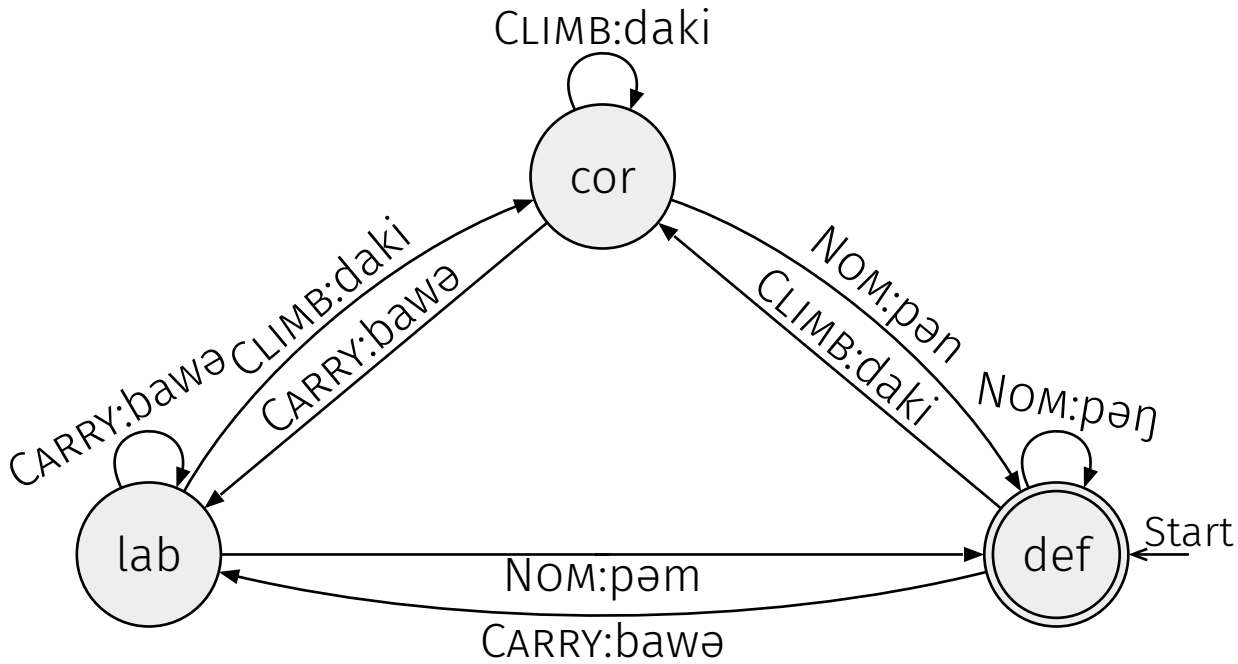


Finite-state machine for λ



Finite-state machine for φ

A proposal



FST for $\varphi \circ \lambda$

A proposal

Hua & Jardine (2021), Hua et al. (2021)

- FST for $\varphi \circ \lambda$ is learnable from known techniques (Oncina et al., 1993)
- Learner then **decomposes** FST for $\varphi \circ \lambda$ into FSTs for φ and λ
- Learner **guaranteed** to identify φ and λ when:
  - λ is homomorphism
  - φ is ISL_2 and represents *one* process
  - Learner sees sufficient combinations of morphemes

A proposal

Chandlee & Jardine (2025)

- **Natural classes** $\rightarrow \varphi$ can have multiple processes
- **Domain information** (Oncina & Varò, 1996) $\rightarrow$ we can learn from data with morphotactic resetrictions

A proposal

Chandlee & Jardine (2025)

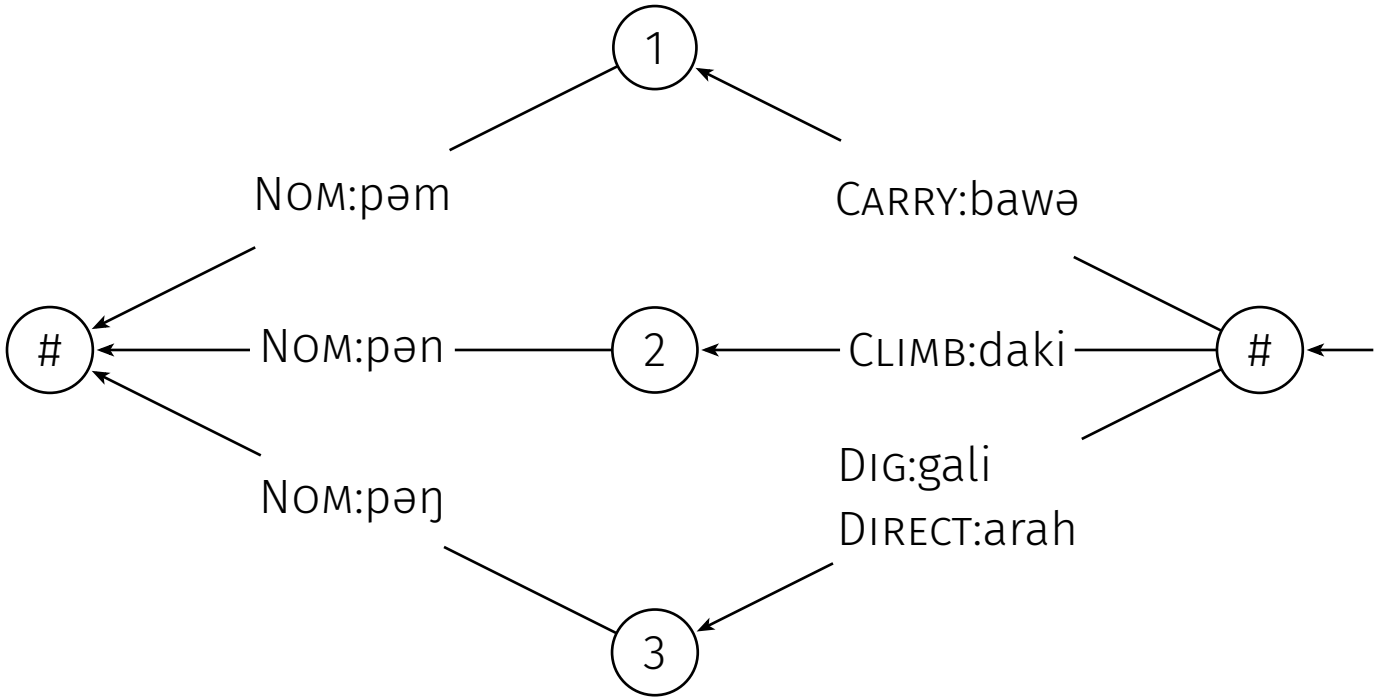
- **Johor Malay** (Onn, 1976)
  - Nasal place assimilation before obs. /pəŋboroŋ/ ↦ [pəmboroŋ] ‘wholesaler’
  - Post-nasal deletion of voiceless obs. /pəŋtali/ ↦ [pən_ali] ‘dancer’
  - Coda /r/ deletion /kisar/ ↦ [kisa_] ‘to answer’
  - Final /a/ raising /bawa/ ↦ [bawə] ‘to answer’
  - Coda /k/ glottalization /masak/ ↦ [masaʔ] ‘to cook’
  - (other processes discussed later)

Input data

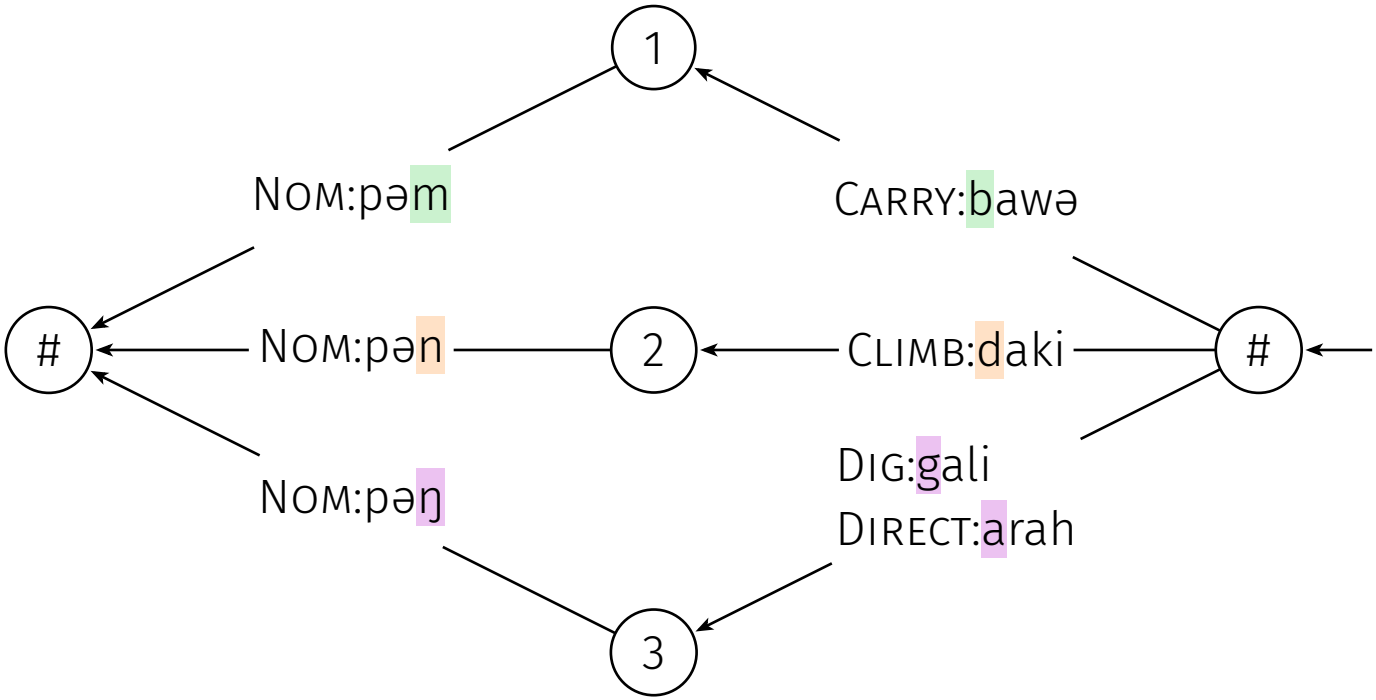
NomDirect	pəŋarah
NomCarry	pəmbawə
NomClimb	pəndaki
NomDig	pəŋgali

A proposal

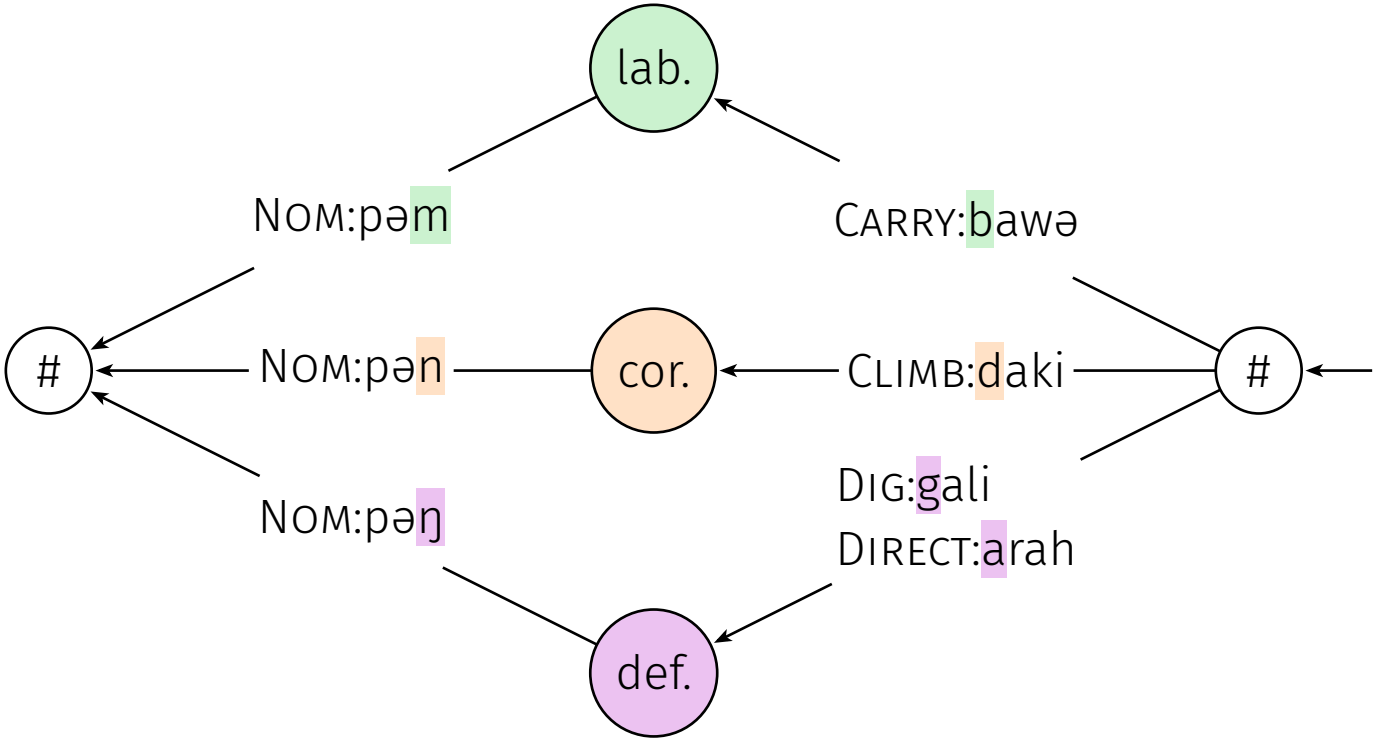
NomDirect	pəŋarah
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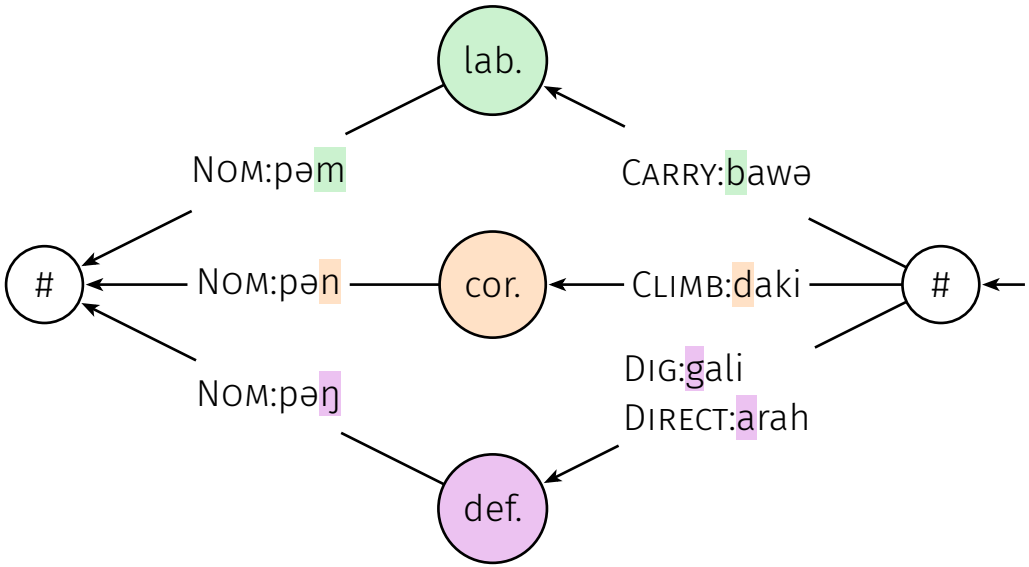
A proposal

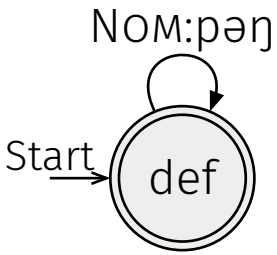

1
NOM:pəm
CARRY:bawə
#
NOM:pən
2
CLIMB:daki
#
NOM:pəŋ
DIG:gali
DIRECT:arah
3

A proposal


lab.
NOM:pəm
CARRY:bawə
#
NOM:pən
cor.
CLIMB:daki
#
NOM:pəŋ
DIG:gali
DIRECT:arah
def.

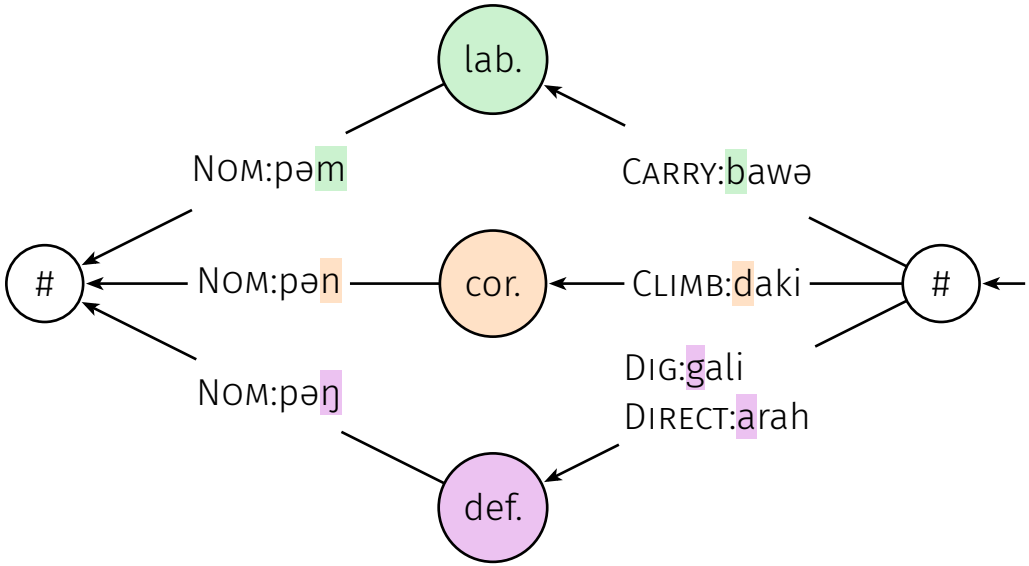
A proposal

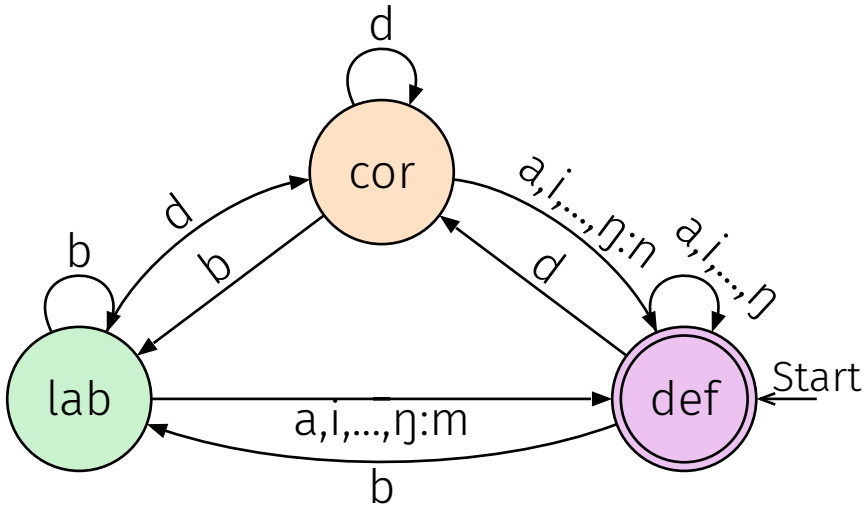




λ FST

A proposal





φ FST

A proposal

Chandlee & Jardine (2025): Dataset from Johore Malay (Onn, 1976)

- **Input**: 14 stems × 5 affixes:

ACT	/məŋ-/
NOM	/pəŋ-/
CAUSE	/-i/
BEN	/-kan/
NOMSUF	/-an/

- **Output**: Correct lexicon and grammar representing all five processes
  - This includes counter-feeding of /r/-deletion and /a/-raising
    /bakar/ ‘to burn’ ↦ [baka] *[bakə]

A proposal

- **Inductive principles**:
  - Phonological functions are **input-local**
  - Phonological functions refer to **natural classes** (in the environment)
  - Error-checking to discover the coda generalizations

A proposal

Limitations

- Bidirectional processes:
  - Nasal deletion before sonorants /pəŋrayu/ ↦ [pə_rayu] ‘appeal’
  - Epenthesis for vowel hiatus /bantuan/ ↦ [bantuwan] ‘aid, relief’

A proposal

- **Other solutions**
  - Constraint-based grammars (Tesar 2014; Jarosz 2006, 2013; Wang and Hayes 2022)
  - Minimum-description length (Rasin et al., 2020, 2021)
  - Bayesian approaches (Barke et al., 2019; Ellis et al. 2022)
- All require some **restriction** on hypothesis space
- Our learner is:
  - Based on *provably correct* algorithms
  - Efficient

Ongoing work

Ongoing work

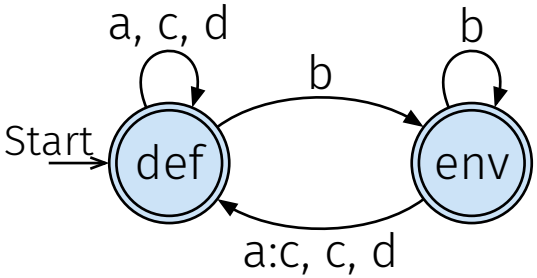
- Interacting processes (Wang)
- Two-sided contexts (Anthony, Jeyaram, Kandepalli)
- Long-distance processes (Czarnecki, Franz)
- Abstract lexical entries (Franz)
- Non-concatenative morphology (Helfand)
- Testing on large corpus of datasets (Heinz et al.)
- ...

Learning Rule Interactions

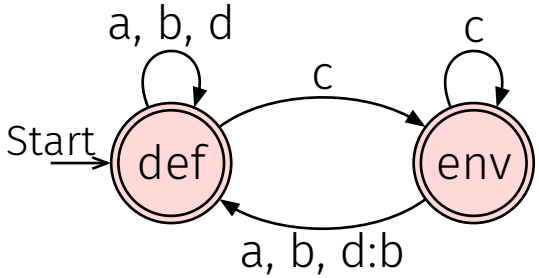
(Wang and Jardine, 2026)



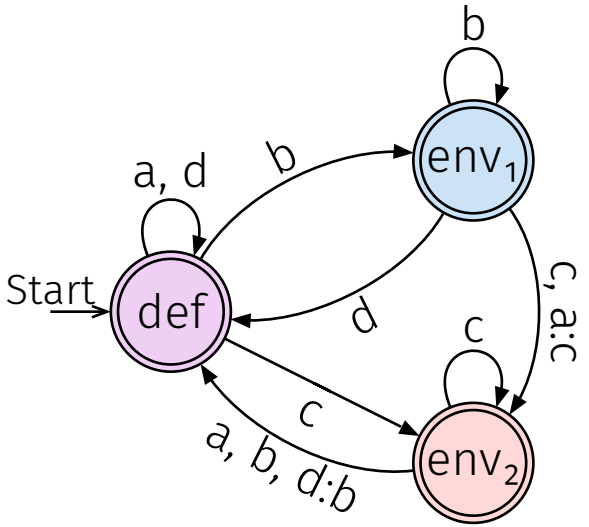
Chenli Wang



a → c / b ___



d → b / c ___



a → c / b ___ > d → b / c ___

Ongoing work

Tiers and UR learning

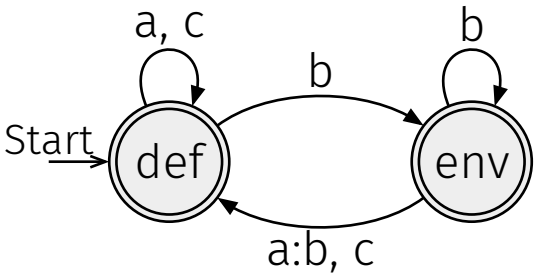
(in prep)



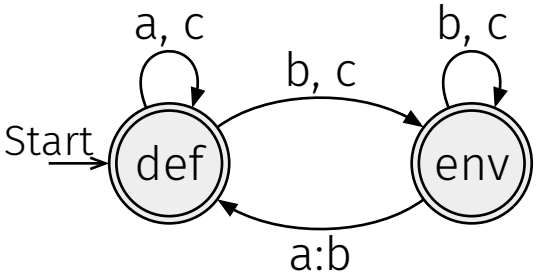
Vincent Czarnecki



Eli Franz



a → b / b ___

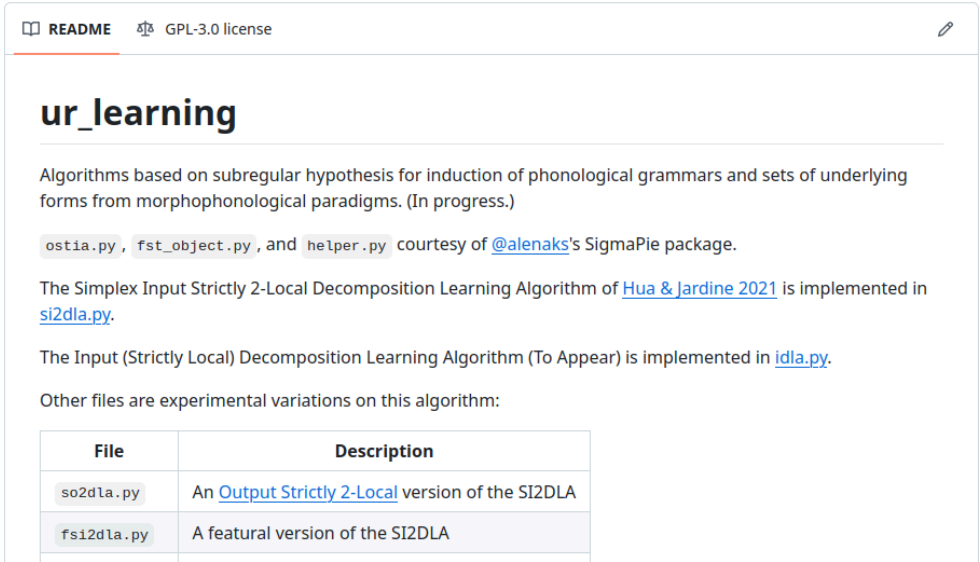


a → b / bc* ___

Implementation with graphical user interface

Neha Jeyaram
Manish Kandepalli
Ezra Anthony

Give it a spin!



README GPL-3.0 license

ur_learning

Algorithms based on subregular hypothesis for induction of phonological grammars and sets of underlying forms from morphophonological paradigms. (In progress.)

`ostia.py`, `fst_object.py`, and `helper.py` courtesy of @alenaks's SigmaPie package.

The Simplex Input Strictly 2-Local Decomposition Learning Algorithm of Hua & Jardine 2021 is implemented in si2dla.py.

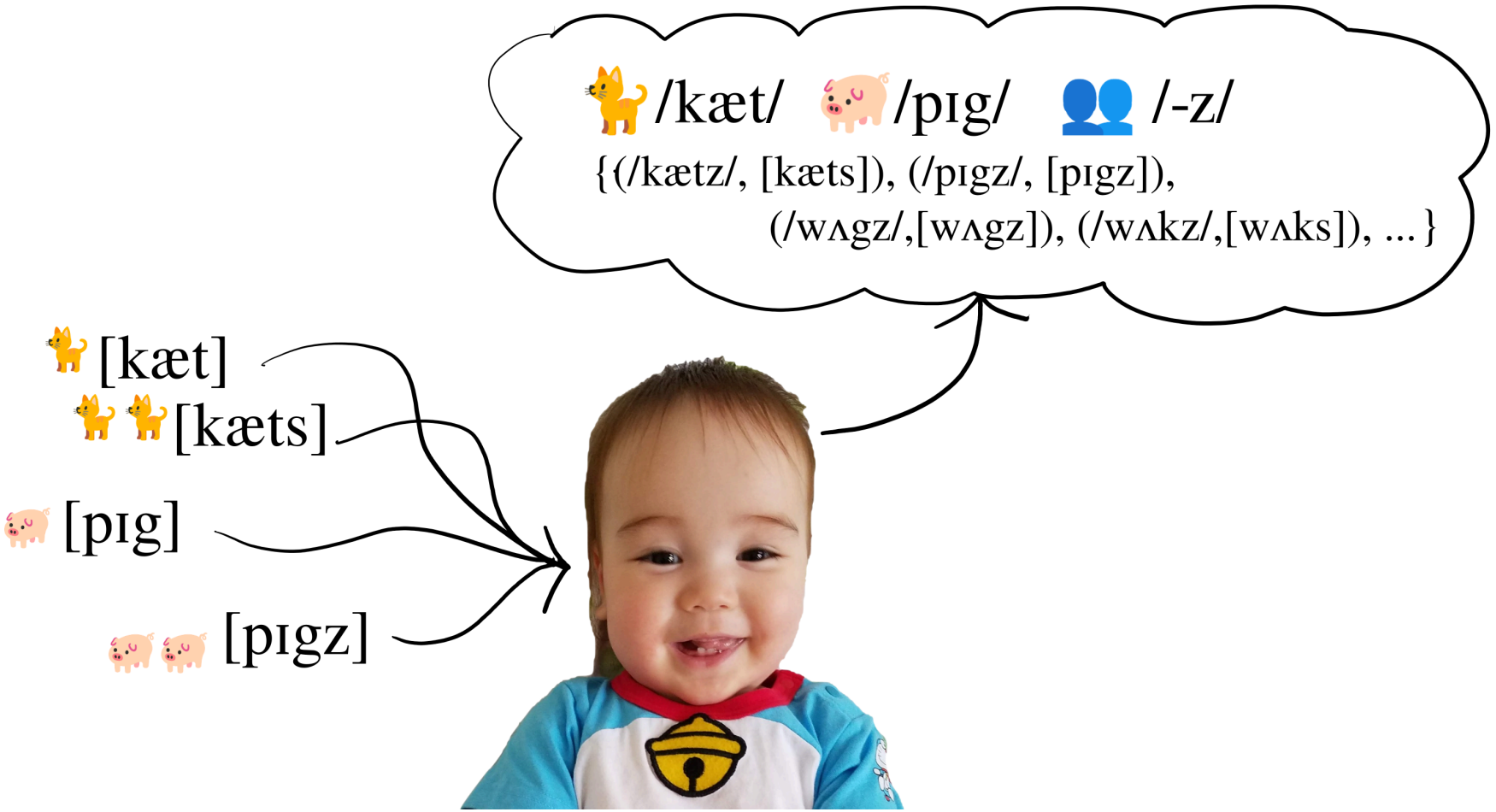
The Input (Strictly Local) Decomposition Learning Algorithm (To Appear) is implemented in idla.py.

Other files are experimental variations on this algorithm:

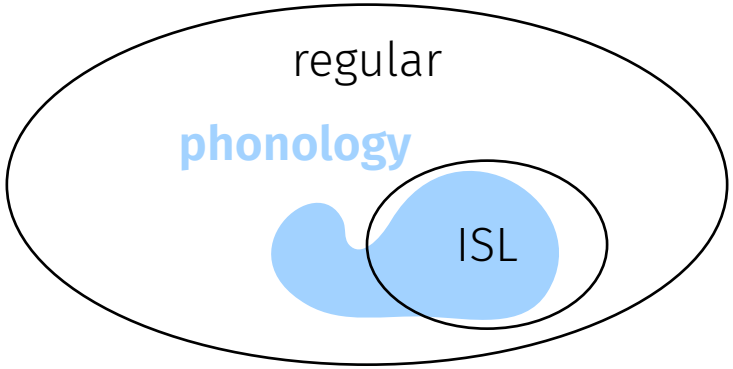
File	Description
`so2dla.py`	An Output Strictly 2-Local version of the SI2DLA
`fsi2dla.py`	A featural version of the SI2DLA

To conclude

To conclude


/kæt/ /pɪg/ /-z/
{(/kætz/, [kæts]), (/pɪgz/, [pɪgz]),
(/wʌgz/,[wʌgz]), (/wʌkz/,[wʌks]), ... }
[kæt]
[kæts]
[pɪg]
[pɪgz]

To conclude



Thank you!